

DATA SCIENCE LEARNING NOTES

DAY 02 • CONDITIONS IN PYTHON

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Goal: Data Scientist

Program: 60-Day Intensive Data Science Journey

Day 2 Overview

Today you learned how Python makes decisions using conditions. These concepts are essential for validation, classification, data processing, and analytical rules.

Learning Flow

if → else → elif → comparison → logical operators → nested conditions → practical programs

1. if Statement

Runs a block when a condition is True.

```
age = 22

if age >= 18:
    print("You are an adult")
```

2. if and else

else runs when the if condition is False.

```
marks = 75

if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

3. Comparison Operators

Comparison operators return True or False.

```
== Equal
!= Not equal
> Greater than
< Less than
>= Greater than or equal
<= Less than or equal
```

4. elif

Checks additional conditions when earlier conditions are False.

```
marks = 75

if marks >= 80:
    print("A")
elif marks >= 70:
    print("B")
elif marks >= 60:
    print("C")
elif marks >= 50:
    print("D")
else:
    print("F")
```

5. Multiple Conditions

Use logical operators to combine comparisons.

```
marks = 85
```

```
if marks >= 80 and marks <= 100:
    print("Excellent")
else:
    print("Other grade")
```

6. Logical Operators

and requires both conditions. or requires at least one. not reverses the result.

```
age = 22
cgpa = 3.23

if age >= 18 and cgpa >= 3.0:
    print("Eligible")

marks = 85

if marks >= 80 or marks == 50:
    print("Condition matched")

is_student = True

if not is_student:
    print("Not a student")
else:
    print("Student")
```

7. Nested Conditions

A nested condition places one condition inside another.

```
age = 22
cgpa = 3.23

if age >= 18:
    if cgpa >= 3.0:
        print("Eligible")
    else:
        print("CGPA is too low")
else:
    print("Age is too low")
```

8. User Input with Conditions

input() returns text. Convert numeric input using int() or float().

```
marks = int(input("Enter your marks: "))

if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

9. Grade Calculator

Multiple conditions can classify marks into grades.

```
marks = int(input("Enter your marks: "))

if marks >= 80:
    grade = "A"
elif marks >= 70:
    grade = "B"
elif marks >= 60:
    grade = "C"
elif marks >= 50:
    grade = "D"
else:
    grade = "F"

print("Grade:", grade)
```

10. Student Performance Analyzer

Combines input, conditions, variables, and output.

```
name = input("Enter student name: ")
marks = int(input("Enter marks: "))

if marks >= 80:
    grade = "A"
    performance = "Excellent"
elif marks >= 70:
    grade = "B"
    performance = "Good"
elif marks >= 60:
    grade = "C"
    performance = "Average"
elif marks >= 50:
    grade = "D"
    performance = "Pass"
else:
    grade = "F"
    performance = "Needs Improvement"

print("\nStudent:", name)
print("Marks:", marks)
print("Grade:", grade)
print("Performance:", performance)
```

11. CGPA Eligibility Checker

Conditions can classify academic performance.

```
cgpa = float(input("Enter your CGPA: "))

if cgpa >= 3.0:
    print("Good academic standing")
elif cgpa >= 2.5:
    print("Needs improvement")
else:
    print("Academic performance needs serious improvement")
```

DAY 2 PRACTICE

Practice 1: Take age and display Adult for 18+, Teenager for 13 to 17, and Child below 13.

Practice 2: Create a grading program for A+, A, B, C, D, and F. Handle invalid marks such as 105.

Practice 3: Take Python, SQL, and Statistics marks. Calculate average and classify the student.

DAY 2 MINI PROJECT

Build a Student Grade Analyzer. Take student name and three subject marks. Calculate total, average, grade, pass/fail status, and performance.

```
name = input("Enter student name: ")
python_marks = int(input("Enter Python marks: "))
sql_marks = int(input("Enter SQL marks: "))
statistics_marks = int(input("Enter Statistics marks: "))

total = python_marks + sql_marks + statistics_marks
average = total / 3

if average >= 80:
    grade = "A"
    performance = "Excellent"
elif average >= 70:
    grade = "B"
    performance = "Good"
elif average >= 60:
    grade = "C"
    performance = "Average"
elif average >= 50:
    grade = "D"
    performance = "Pass"
else:
    grade = "F"
    performance = "Needs Improvement"

print("\nStudent:", name)
print("Total:", total)
print("Average:", average)
print("Grade:", grade)
print("Performance:", performance)
```

QUICK REVISION

Concept	Purpose
if	Runs code when a condition is True.
elif	Checks another condition.
else	Runs when previous conditions are False.
==	Equal comparison.
!=	Not equal comparison.
>, <	Greater than, less than.
>=, <=	Greater or less than or equal.
and	Both conditions must be True.
or	At least one condition must be True.
not	Reverses a Boolean result.

Concept	Purpose
=	Assigns a value.
int()	Converts input to integer.
float()	Converts input to decimal.

DAY 2 COMPLETE

Day 2 is complete. Next is Day 3: Loops, including for, while, range(), break, continue, nested loops, and practical data-processing exercises.

Muhammad Ahmad • 60 Days Data Science Journey